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Advanced machine learning techniques for coronary artery disease risk prediction in early middle-aged adults

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Abstract

Background Coronary artery disease (CAD) is the leading cause of death globally, with early risk prediction being vital for timely intervention. In this study we utilized advanced machine learning techniques, incorporating clinical, lifestyle, and Genetic factors with the aim to enhance the 10-year risk prediction of CAD in early middle-aged adults (40–55 years).

Methods For developing the machine learning models, we used data from the UK Biobank, a large cohort of over 500,000 participants. Among participants aged 40 to 55, we identified 3,012 CAD cases and 155,176 controls, without any prior CAD diagnosis. More than 700 clinical and lifestyle variables were evaluated, alongside a polygenic risk score (PRS) derived from more than 560,000 genetic variants. Forward feature selection was performed, followed by the application of a stacking ensemble model combining Random Forest and Deep Learning Neural Networks as base learners with a Logistic Regression model serving as a meta-learner. The model's performance for the 10-years prediction of CAD risk was evaluated using Area Under the Receiver Operating Characteristic Curve (AUC) and calibrated against the ACC/AHA guidelines.

Results Forward feature selection identified 22 CAD risk factors, including the PRS. The final model yielded an AUC of 0.81 in the training dataset and 0.80 in the testing dataset. At the 7.5% risk threshold, the model demonstrated higher sensitivity (0.511 vs. 0.346) and comparable specificity (0.863 vs. 0.866) compared to the ACC ASCVD model. The Net Reclassification Index (NRI) improved by 13.79%, emphasizing the model's accuracy and alignment.

Conclusions Our findings highlight the potential of combining Genetic information with clinical and lifestyle factors through machine learning to improve CAD risk prediction in early middle-aged adults. The use of a stacking ensemble approach enabled modeling of complex relationships, yielding superior performance over standard risk models. This framework supports more precise and actionable 10-year risk stratification, reinforcing the value of PRSs when combined with traditional factors. Future research should validate these methods across diverse populations and clinical environments.

Keywords Coronary artery disease (CAD), Machine learning, Risk prediction, Polygenic risk scores (PRS)

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Introduction

Coronary artery disease (CAD) is the leading cause of mortality worldwide [1], and early detection of individuals at high risk is crucial for the clinicals with the intention of guiding them to a healthier lifestyle and proper medication [2, 3]. The development of CAD is a complex process that involves an interplay among clinical, lifestyle and genetic factors [4].

Clinical and lifestyle factors can enlighten the screening and comprise an important role in the prediction of CAD, including age, sex, race and ethnicity, smoking habits, total cholesterol, low-density and high-density lipoprotein cholesterol, systolic blood pressure, diabetes, use of medications, blood biochemistry measurements, physical activity related parameters and mental health related parameters. Risk prediction models have been developed and widely used for both 10-year and lifetime risk prediction of CAD [5, 6], embracing these factors, including the Framingham Risk Score [7], the Atherosclerotic Cardiovascular Disease (ASCVD) Risk Calculator, proposed by the American College of Cardiology (ACC)/American Heart Association (AHA) Guidelines [8] and the QRISK score [9]. Moreover, over the last decade, the addition of genetic factors, in the light of polygenic risk scores (PRSs), to those risk models, have shown improved performance and accuracy on the CAD risk prediction [10, 11].

Lately, machine learning models have been proposed aiming to improve the accuracy of risk prediction models for several diseases [12–14]. Recent studies have demonstrated how artificial intelligence and integrated machine learning pipelines can significantly enhance the prediction of complex disease outcomes across various medical domains, such as oncology and immunotherapy [14–16].

Regarding CAD, the contribution of machine learning techniques has been assessed in several studies, with the goal of improving the accuracy of clinical diagnosis or prediction of diseases [17–22]. In today's medical landscape, there are numerous modern diagnostic tests available, and when paired with machine learning, these tests become powerful tools for cardiologists and can play a crucial role in both preventing and diagnosing coronary artery disease [23].

Despite these advances, early middle-aged adults remain significantly underrepresented in the derivation and validation cohorts of most cardiovascular risk models [24, 25]. Models such as the ASCVD Risk Calculator [8] and the Framingham Risk Score [26] were primarily developed in middle-aged and older populations, resulting in decreased predictive accuracy among younger subgroups. Recent efforts, such as QRISK3-Young [27] and PREVENT [28], have aimed to address this gap by tailoring prediction tools to younger populations. However, these models still rely predominantly on traditional clinical risk factors and do not fully account for the complex interplay between genetic predisposition and environmental exposures that contribute to early-onset CAD [29]. Furthermore, the integration of polygenic risk scores and lifestyle factors into predictive modeling for early middle-aged adults remains limited, underscoring a significant methodological gap [30].

In the present study, we integrated a large number of clinical and lifestyle factors combined with genetic risk, using aggregated and composite machine learning models, developed and validated on a large biobank with ~500,000 participants. Our main target was to assess the 10-year risk prediction of CAD in early middle-aged adults, aged 40 to 55 years, a population in which CAD often progresses more aggressively, as individuals in this age group with premature atherosclerosis have been shown to experience higher relative risks and worse outcomes compared to older individuals [31, 32]. Furthermore, this age group presents a key period where the initiation of statin therapy and lifestyle modifications, such as improved diet, increased physical activity, and smoking cessation, can yield significant benefits in both reducing disease progression and preventing adverse cardiovascular events [33, 34]. Finally, the predictive performance of our model was systematically evaluated against the well-established and widely utilized CAD risk prediction model developed by the ACC [8].

Methods

Study participants

The analysis was performed in the UK Biobank (UKB). UKB data were derived under the Application ID 64,586. UKB is a population-based cohort of more than 500,000 individuals, aged 40–69 years, with rich phenotypic, demographic, clinical and -omics data such as genomics and metabolomics [35]. Participants attended assessment centers at which baseline data were collected on their medical history, lifestyle, body composition and environment. DNA for Genotyping was collected as well. For the validation of the Genetic risk model, we used a subset of unrelated individuals from multiple ethnicities, by removing one of each pair of 1st and 2nd degree relatives, using the centrally provided kinship data from UKB [36].

CAD incidence in the UK biobank

Data on CAD incidence in UK Biobank were available for each participant based on measurements and self-reported data collected during their first visit and, longitudinal general practice (GP) data including medications, clinical data such as measurements and diagnoses made prior to the 1st visit and, in addition, data from reports derived from Hospital Episodes Statistics. We focused on early prevention in early middle-aged adults. We identified 3012 CAD cases, with a CAD diagnosis according

to ICD-10 (I25.1) code or a death caused by CAD within a period of 10 years counted from the patient's first visit time registered in the UKB. Patients with an age more than 55 years at baseline date were excluded from the analysis. Patients with CAD diagnosis (I25.1), MI diagnosis (I21.0, I21.1, I21.2, I21.3, I21.4, I21.9, I22.0, I22.1, I22.8, I22.9, I23.0, I23.1, I23.2, I23.3, I23.5, I23.6, I23.8, I25.2) or stroke (I63.0, I63.1, I63.2, I63.3, I63.4, I63.5, I63.6, I63.8, I63.9) prior to the baseline date or up to 3 months after the baseline date were also excluded. We moreover selected 155,176 controls, aged less than 55 Years at baseline date, that were not diagnosed for CAD in a period of 10 Years from the baseline date. We therefore included 158,188 participants in our analysis.

Clinical and lifestyle features included

To develop our prediction model we examined a list of more than 700 clinical and lifestyle factors including blood biochemistry (e.g. HDL, LDL, A1c and many more), blood count biomarkers (e.g. red and white cell count, basophil count and many more), blood pressure measurements (SBP, DBP), body size measurements, smoking status at baseline date, medications used on baseline date, diagnoses made prior to baseline date, physical activity related self-reported parameters, mental health related self-reported parameters and sleep related self-reported parameters. Participants with missing HDL measurement at baseline date were excluded from the analysis.

Polygenic risk score

On top of the factors for our prediction model, we included the Polygenic Risk Score (PRS). PRS represents a sum across single-nucleotide proteins (SNPs) of the number of effect alleles at that SNP, weighted by their effect size estimates (beta coefficients). For our developed PRS, we used the beta coefficients from a large independent multiethnic GWAS meta-analysis of ~9.4 million Genetic variants, including 68 K CAD cases and 117 K controls [37]. For the PRS calculation, we used only unrelated participants. For each individual i in the UK Biobank the PRS was calculated as:

$$PRS_i = \sum_{j=1}^{K} b_j \cdot g_{ij}$$

where b_j is the beta coefficient for the SNP j, g_{ij} is the genotypic value for SNP j for individual i (0/1/2, depending on the number of the risk alleles for the certain SNP). We checked various p-value thresholds (5 × 10e-8, 0.01, 0.1 and 0.5) for the inclusion of the variants in our PRS, with the aim to optimize its predictive ability. Genetic principal components (PCs) to account for population stratification were centrally computed, in order to adjust for the several ethnicities included in our dataset. We used PRS-CS software [38] for Linkage Disequilibrium (LD), using the p-value threshold of 0.5 and the defaults in all the optional parameters. The PRS was standardized per unit increase.

Development of the machine learning models

To build the machine learning models, the dataset of 158,188 participants was split randomly into two halves, a training and a testing dataset, each containing 1,506 CAD cases and 77,588 healthy controls respectively.

The whole process started with feature selection, performed in the training dataset with more than 700 factors examined, including the PRS and the PCs. For each pair of highly correlated factors (Pearson's correlation coefficient > 0.9), the one demonstrating the higher discriminative ability, as measured by the Area Under the Receiver Operating Characteristic Curve (AUC) derived from a logistic regression model with CAD events as the dependent variable, was retained. Continuous factors were standardized to have zero mean and unit variance, and categorical/binary factors were transformed using one-hot encoding prior to model training. Binary factors with the less common value rate < 1% were also removed. Missing data for a given factor were imputed using median values when the proportion of missing values was ≤ 40%; otherwise, the factor was excluded from the analysis.

A forward feature selection was performed with criteria of AUC increase of at least 0.05% when running a logistic regression over selected factors and the CAD phenotype (ICD10 code I25.1) as the dependent column. LDL was added to the feature list to allow monitoring and manipulating its effect on the risk. Diabetes diagnosis replaced the original elected A1c levels as the AUC difference was negligible between the two and diabetes diagnosis is more commonly available for individuals both in the dataset and in real life applications.

We then applied several aggregated machine learning techniques. A stacking Ensemble Learning model was then built, composed of two base models - a Random Forest model and a Deep Learning Neural Network model with the rationale of capturing nonlinear and complex relationships between features for prediction improvement, and a Logistic Regression model as meta-learner, accepting as input the probabilities derived from the base models as well as the selected features.

Hyperparameter tuning for each model was performed using a random search strategy. For the Random Forest model, the optimal hyperparameters were determined as follows: 500 trees, maximum tree depth of 10, minimum number of samples required to split an internal node set to 2, minimum number of samples required at a leaf node set to 10, maximum number of features considered

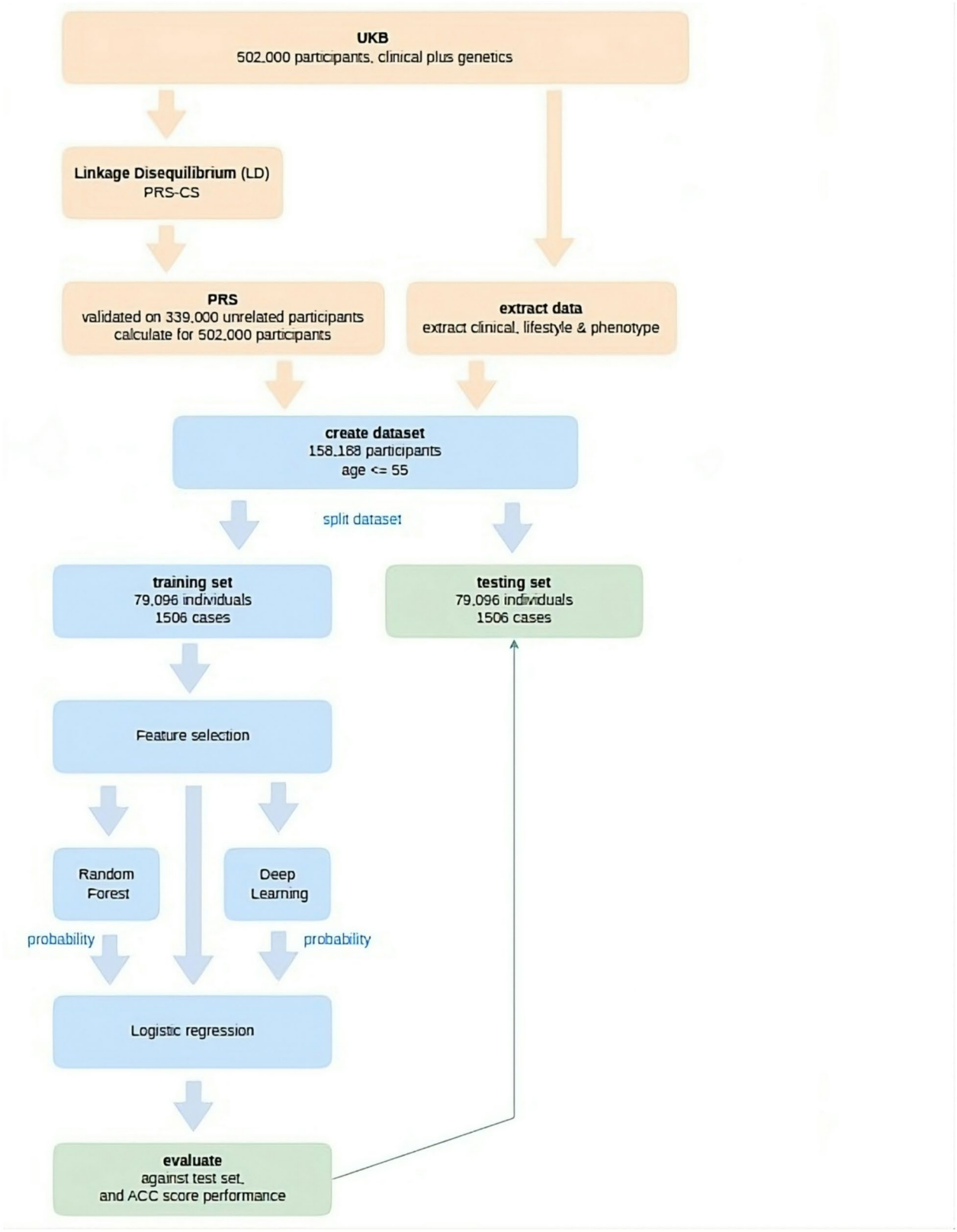


Fig. 1 Schematic representation of the study

at each split set to the square root of the total number, entropy criterion for information gain, and bootstrap sampling enabled.

For the Deep Learning Neural Network, the optimal architecture included three hidden layers comprising 53, 41, and 57 nodes respectively, with the hyperbolic tangent (tanh) used as the activation function and trained over 8 epochs. Probability outputs from the two base models were combined with the independent variables derived from the forward feature selection. To construct the meta-learner logistic regression classifier, features with p-values greater than 0.05 were excluded. For the logistic regression model, hyperparameter tuning selected a regularization strength (C) of 0.021, an L2 penalty, and the "saga" solver for optimization. The AUCs of the base models and the final model were assessed on the training set using 20-fold cross-validation while performance on the test set was assessed using 100 bootstrap iterations.

For each participant, the risk was calculated using the following formula:

$$Risk = 1 - SR^{e^{(S-meanS))}} \quad (1)$$

In Eq. (1)

- *SR* represents the survival rate, which is calculated separately for men and women and applied accordingly to each individual. The survival rate is defined as.

$$SR = \frac{T - N}{T} \quad (2)$$

where T is the total number of records (per gender), and N is the number of events (per gender).

- S is the linear combination of coefficients and feature values, including a constant term, as expressed, as shown in Eq. (3)

$$S = \Sigma Ci \cdot Vi + constant \quad (3)$$

where C_i is the coefficient of the feature i, and V_i is the corresponding feature value.

- *meanS* is the mean S value among the participants in the dataset, calculated separately for men and women and used respectively for each individual.

Finally, the risk estimates from the logistic regression model were calibrated to match the slope of the ACC/AHH ASCVD 10y risk score and compared to the widely used model recommended by the ACC/AHA Guidelines on the Assessment of Cardiovascular Risk (the ACC ASCVD model). The comparison was based on sensitivity, specificity, and Net Reclassification Indices (NRIs) across several clinically relevant risk thresholds, including the 7.5% statin initiation threshold recommended in current U.S. guidelines [39]. An overview of the modeling procedure is presented in Fig. 1.

Table 1 Selected factors after performing the forward feature selection in the training dataset

Characteristic	(*N* = 79,094)
Sex (Female), *N* (%)	
Females	44,363 (55.28)
Males	35,894 (44.72)
Systolic Blood Pressure (SBP) in mmHg, mean (sd)	130.05 (16.76)
Low-density lipoprotein (LDL) direct in mmol/L, mean (sd)	3.53 (0.82)
Cystatin C in mg/L, mean (sd)	0.85 (0.14)
Lipoprotein A in nmol/L, mean (sd)	44.47 (48.96)
Basophill count in 10^9 cells/L, mean (sd)	0.04 (0.05)
3rd genetic PC, mean (sd)	0.25 (17.84)
Log (age^2), mean (sd)	15.01 (0.72)
Total cholesterol level divided by HDL level in mmol/L, mean (sd)	4.14 (1.18)
Testosterone in nmol/L, mean (sd)	6.46 (6.13)
C-Reative protein in mg/L, mean (sd)	2.32 (3.96)
Mean Corpuscular Hemoglobin in picograms, mean (sd)	31.28 (1.99)
PRS (standardized), mean (sd)	0.002 (1.003)
Usual Walking Pace, N (%)	
Slow	4307 (5.37)
Steady	38,999 (48.59)
Brisk	36,466 (45.41)
Smoking, N (%)	
No	65,533 (81.65)
Yes	14,724 (18.35)
Use of Statins, N (%)	
No	75,298 (93.82)
Yes	4959 (6.18)
Hypertension or hypertensive treatment, N (%)	
No	69,316 (86.37)
Yes	10,941 (13.63)
Asthma (J45.9), N (%)	
No	77,998 (97.19)
Yes	2259 (2.81)
Loneliness Isolation, N (%)	
No	62,368 (77.71)
Yes	16,265 (20.27)
Seen GP for nerves, anxiety, tension or depression, N (%)	
No	51,781 (64.52)
Yes	27,799 (34.64)
Mood swings	
No	51,781 (64.52)
Yes	27,799 (34.64)

Table 2 Predictive performances of the two base models and the final logistic regression model

	AUC (95% CI)	
Model	**Train dataset**	**Test dataset**
Random Forest	0.80 (0.79, 0.81)	0.79 (0.78, 0,80)
Deep Learning Neural Network	0.78 (0.78, 0.80)	0.79 (0.78, 0.80)
Logistic Regression	0.81 (0.80, 0.82)	0.80 (0.79, 0.80)

Table 3 Quantitative measures of calibration

Measures	Train dataset	Test dataset
R^2	0.99	0.99
Slope	0.50	0.50
Intercept	0	0
p-value	1.65e-06	5.78e-06
Brier Score	0.00157	0.00201

Model performance evaluation

To evaluate model performance beyond traditional discrimination metrics, we examined precision-recall curves (PRCs), which are particularly informative in contexts of class imbalance, such as CAD occurring in early middle-aged adults. The PRCs were constructed by plotting precision (positive predictive value) against recall (sensitivity) across a range of decision thresholds. To explore potential variation in model behavior across subgroups, we also stratified PRC analyses by age deciles, providing further insight into the discriminative capacity of the models among younger individuals.

Model calibration was assessed both visually, through calibration plots, and quantitatively, by calculating the calibration slope, intercept, R^2, and the Brier score, separately for the training and testing datasets.

Model explainability and clinical utility

To evaluate the potential clinical benefit of our model, we performed Decision Curve Analysis (DCA). Net benefit was calculated across a range of threshold probabilities (from 0 to 0.20), allowing comparison between our model and the standard ACC/AHA ASCVD risk score. The resulting decision curves were compared to the default strategies of "treat all" and "treat none", to evaluate the practical utility of each model in informing treatment decisions.

In parallel, model interpretability was assessed using SHAP (SHapley Additive exPlanations) values. SHAP analysis was conducted via the LinearExplainer module from the SHAP Python package (version 0.47), producing summary plots illustrating the relative contribution of each feature to the model's output.

Results

Our training dataset in the UK Biobank consisted of 1,506 CAD cases and 77,588 healthy controls, with a mean age of 48.3 Years and 55.35% of participants being female. The forward feature selection process identified 22 factors for CAD risk prediction and achieved an AUC of 0.806. The selected factors included a PRS, consisting of 564,856 variants, at the p-value threshold of < 0.5, the third genetic PC, and clinical, lifestyle and mental health related factors. Details of the selected features are shown in Table 1.



Fig. 2 Calibration of the logistic regression classification model to match ACC ASCVD score

Table 4 Comparison of the predictive performance of the final logistic regression classification model against the ACC ASCVD model

				Logistic Regression Model							ACC ASCVD Model					
Risk level	NRI	NRI events	NRI non-events	Sensitivity	Specificity	PPV	TP	TN	FP	FN	Sensitivity	Specificity	PPV	TP	TN	FP	FN
0.05	9.47%	9.95%	−0.48%	0.677	0.770	0.062	1190	60,426	18,074	568	0.577	0.771	0.054	1015	60,803	17,697	743
0.075	13.79%	16.50%	−2.70%	0.511	0.863	0.077	899	67,730	10,770	869	0.346	0.866	0.066	609	69,852	8648	1149
0.1	14.42%	18.15%	−3.72%	0.392	0.909	0.088	689	71,393	7107	1069	0.210	0.913	0.081	370	74,317	4183	1388
0.2	12.73%	15.02%	−2.29%	0.179	0.973	0.130	315	76,393	2107	1443	0.029	0.974	0.140	51	78,188	312	1707

PPV Positive predictive value, *TP* True positives, *TN* True negatives, *FP* False positives, *FN* False negatives

The stacking ensemble learning model, composed of a Random Forest model and a Deep Learning Neural Network model as base models and a Logistic Regression meta-learner model, was then trained on the training dataset. For the Random Forest model, the AUC was calculated using 20-fold cross-validation, providing values of 0.80 on the training dataset and 0.79 on the testing dataset. Regarding the Deep Learning Neural Network model, the AUC was also calculated using 20-fold cross-validation, yielding values of 0.78 for the training dataset and 0.79 for the testing dataset (Table 2).

Finally, a classification model developed, utilizing a logistic regression framework, incorporating as inputs the factors that were derived from the forward feature selection and the predicted risk probabilities from the Random Forest and Deep Learning Neural Network models. This classification model achieved a predictive performance of AUC = 0.81 for the training dataset and AUC = 0.80 for the testing dataset, using 20-fold cross-validation for training set performance and 100 bootstrap iterations for test set performance evaluation (Table 2). The individual risk values were then calculated for each participant.

Calibration of the model was evaluated both visually, through calibration curves, and quantitatively, through calibration slope, intercept, Brier score, and R^2 metrics. Calibration results for both datasets are presented in Table 3, and calibration curves are depicted in Fig. 2. The slope was adjusted to align with that of the ACC/AHA ASCVD risk score to ensure comparability with clinical practice.

We also compared the results of the calibrated classification model against the ACC ASCVD performance scores at four risk levels: 5%, 7.5%, 10% and 20%. (Table 4). Overall, our classification model demonstrated better performance with higher sensitivity, specificity, positive NRI and better NRI events at all four risk levels. Furthermore, at the clinically significant risk threshold of 7.5% which serves as the provisional statin initiation threshold in current US treatment guidelines, our classification model achieved a sensitivity of 0.511 and a specificity of 0.863, compared to a sensitivity of 0.346 and a specificity of 0.866 of the ACC ASCVD model. At the same risk level, the NRI of our model increased by 13.79%, and the NRI events improved by 16.50% compared to the ACC ASCVD model (Table 4).

In addition to AUC comparisons, model performance was evaluated using precision-recall curves (PRCs), to account for the imbalanced nature of the dataset. Our model demonstrated superior precision across a range of recall values compared to the ACC/AHA score and the individual base models (Random Forest and Neural Network), with visual reference to key clinical precision thresholds (Fig. 3).

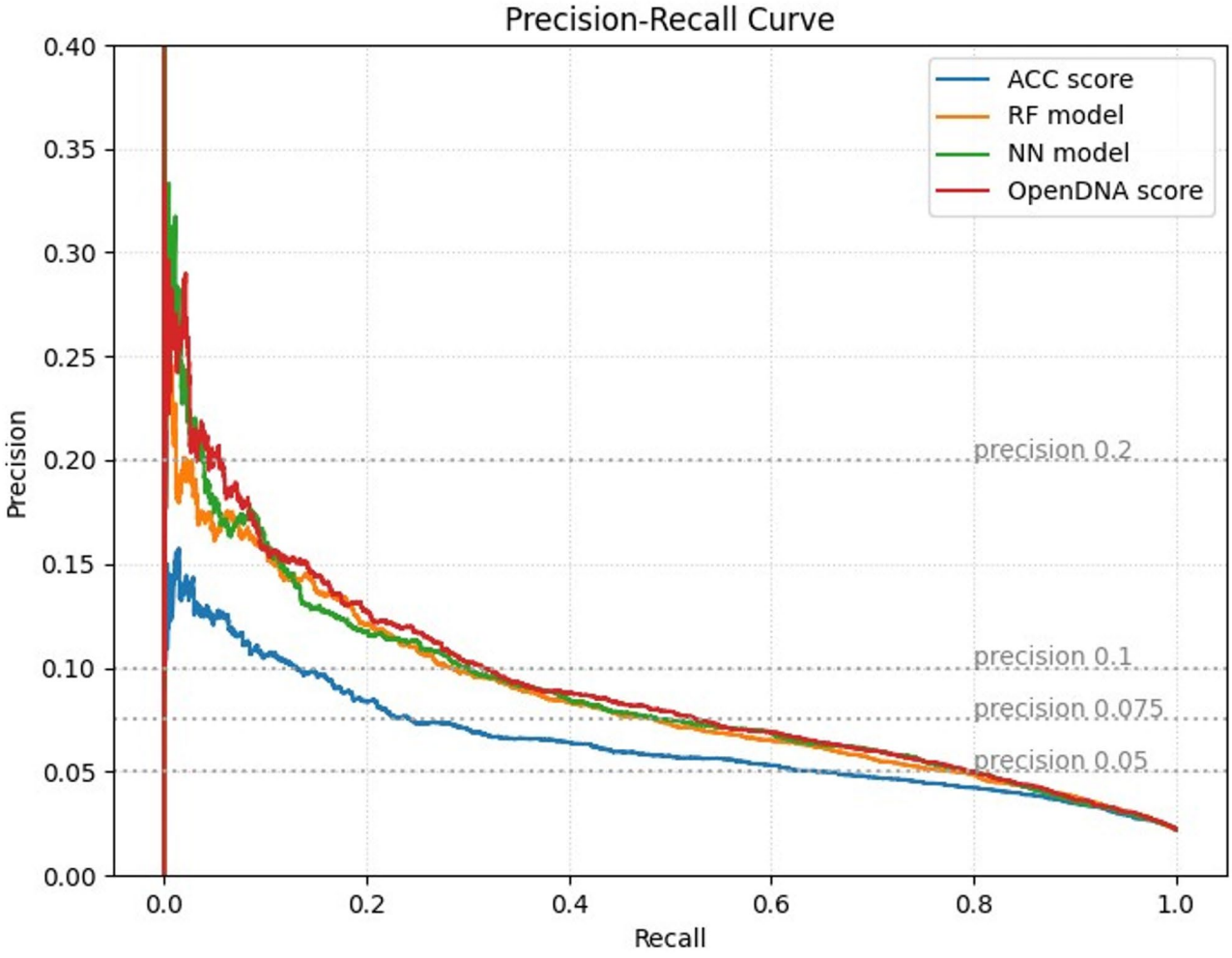


Fig. 3 Precision-recall curve (PRC) comparison across models ACC score: ACC/AHA ASVCD 10y risk score, RF model: random forest model, NN model: Neuron Network model, OpenDNA score: OpenDNA 10y risk score

To further examine model behavior across age subgroups, PRC analyses were stratified by age deciles. In most age strata, our model demonstrated higher recall at fixed precision levels compared to the ACC/AHA model, demonstrating consistent robustness among early middle-aged adults (Fig. 4).

Moreover, DCA showed that our model yielded consistently greater or equal net benefit compared to the ACC/AHA ASCVD risk score across a broad range of threshold probabilities, particularly below the 7.5% clinical intervention threshold (Fig. 5). Finally, SHAP analysis indicated that the top contributors to model predictions were sex, age, and the CAD polygenic risk score, followed by key lifestyle and metabolic factors (Fig. 6).

Discussion

In this study, we conducted a comprehensive evaluation of various clinical and lifestyle factors, combined with Genetic risk, using aggregated and composite machine learning techniques, to enhance 10-year CAD risk prediction in early middle-aged adults. The analysis was conducted on a vast dataset obtained from a large biobank, with more than 150,000 participants, and our model achieved an AUC of 0.80 and outperformed the ACC ASCVD model across multiple risk levels.

Our study focused on early middle-aged adults, where CAD progresses more aggressively and early prediction is critical to prevent severe consequences [40, 41]. Moreover, timely identification allows for effective interventions [42] and preventive strategies, including statins, can alter the disease trajectory and have been shown to be more effective in this age group, yielding better long-term cardiovascular outcomes [43].

Several previous studies have utilized UKB data to develop machine learning models for CAD risk prediction [44–46]. However, our study focused exclusively on individuals under 55 Years old, addressing the need for early detection, and utilized a PRS comprised of more than 560,000 variants enhancing the precision of genetic assessment. Furthermore, the implementation

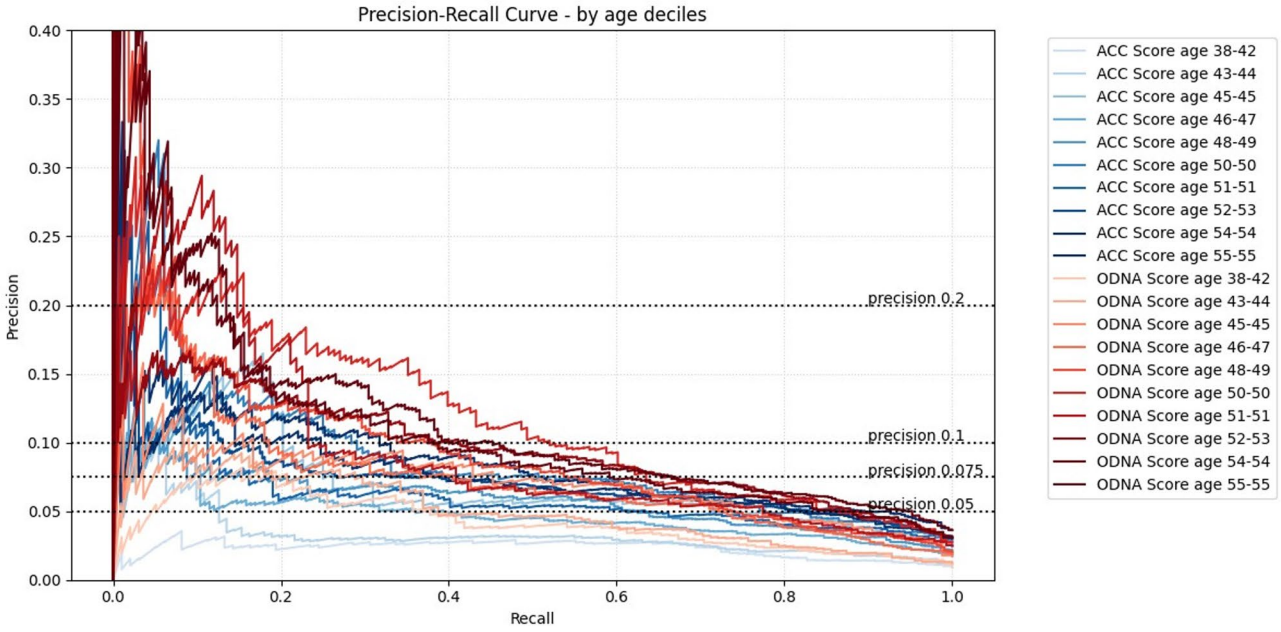


Fig. 4 Precision-recall curves (PRCs) stratified by age deciles ACC score: ACC/AHA ASVCD 10y risk score, ODNA score: OpenDNA 10y risk score

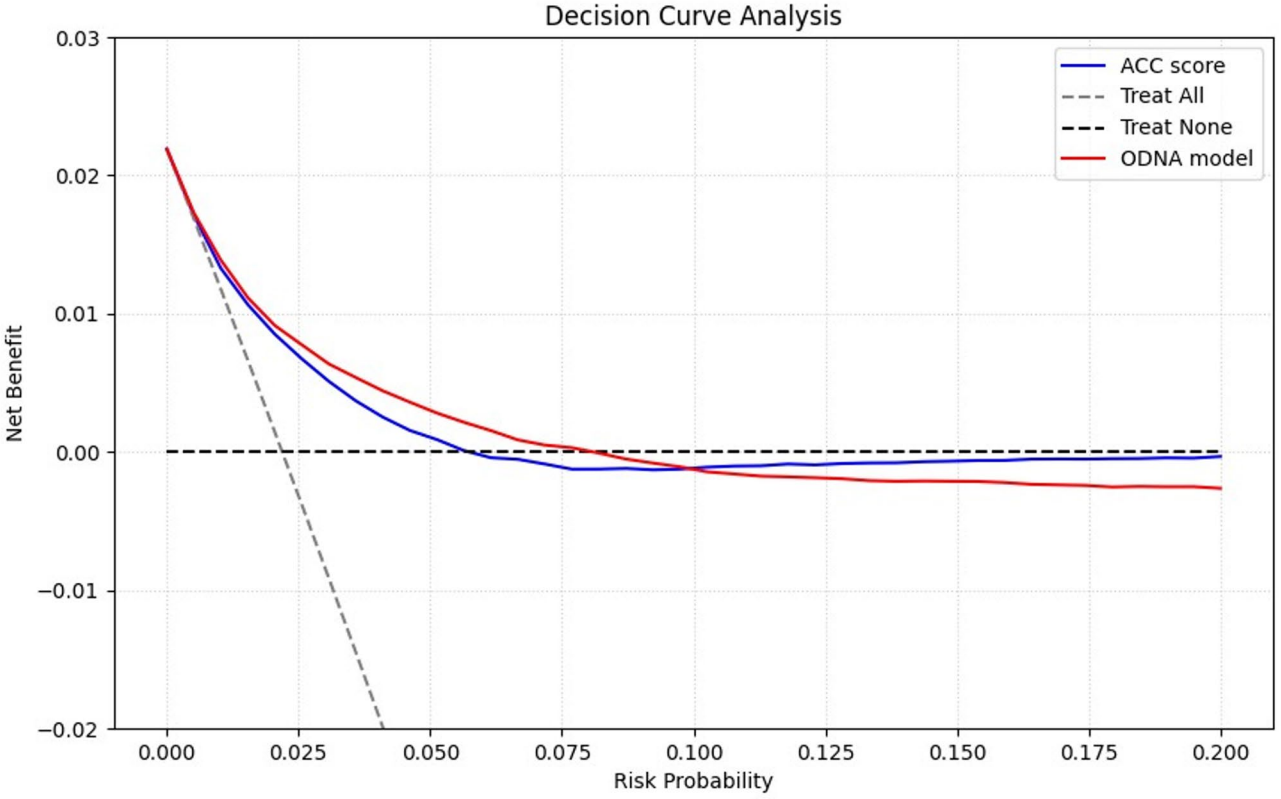


Fig. 5 Decision Curve Analysis (DCA) comparing our model and the ACC/AHA ASCVD risk score ACC score: ACC/AHA ASVCD 10y risk score, ODNA score: OpenDNA 10y risk score

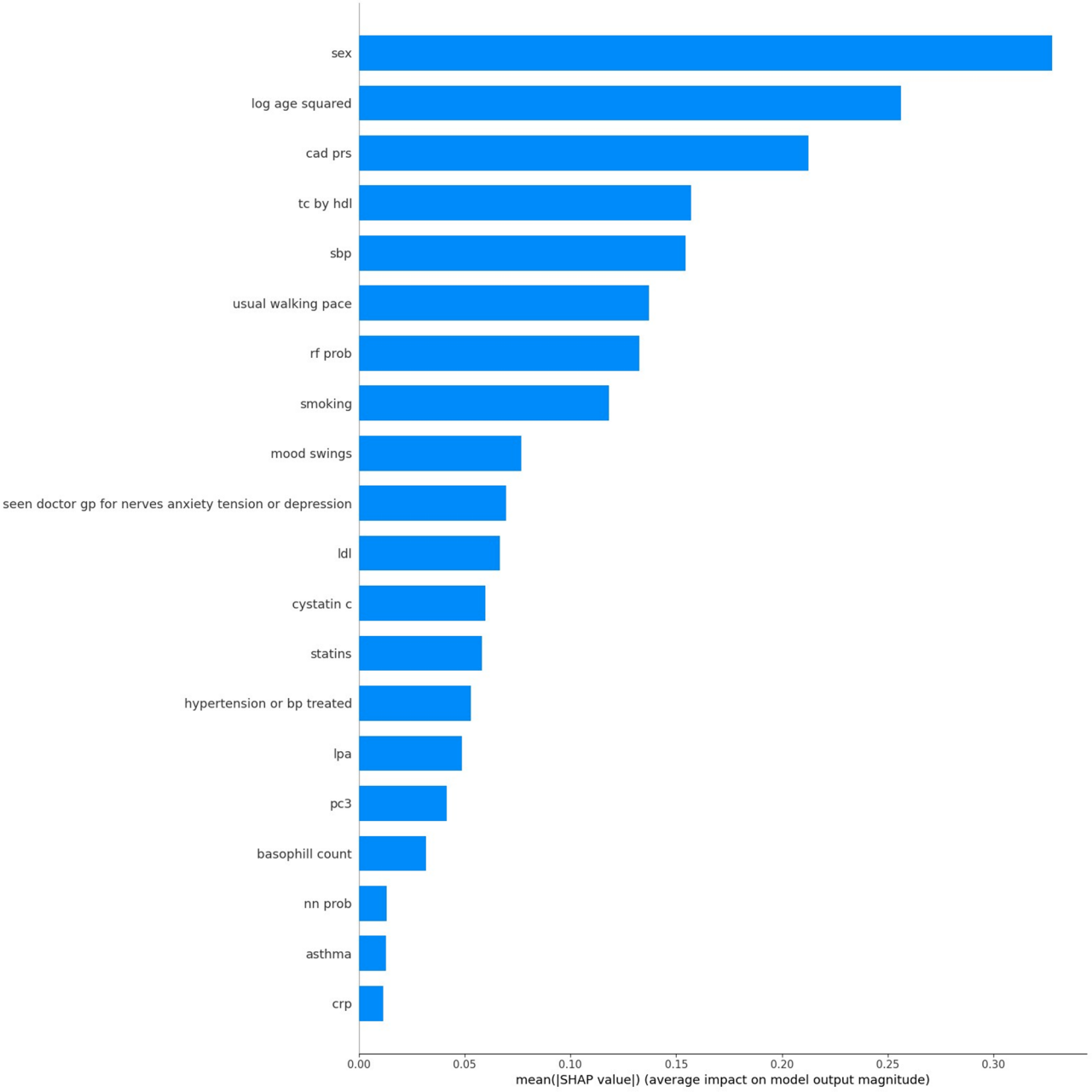


Fig. 6 SHAP summary plot showing feature importance in the final prediction model tc by hdl: total cholesterol divided by high-density lipoprotein, rf prob: random forest model probability outcome, nn prob: neuron network model probability outcome. sbp: systolic blood pressure. crp: C-reactive protein. ldl: low-density lipoprotein. hypertension: diagnosis of hypertension. bp treated: anti-hypertension treatment. statins: statins treatment. lpa: lipoprotein(a), pc3: the third principal component. asthma: diagnosis of asthma

of a stacking ensemble model combining Random Forest and Deep Learning Neural Network techniques provided enhanced predictive accuracy. Finally, in our study we calibrated our model to align with the ACC/AHA guidelines, ensuring enhanced clinical relevance and practical applicability in real-world settings.

Our research highlights several notable strengths that enhance robustness and practical significance in a clinical context. First, the use of UKB, encompassing over 500,000 participants with rich phenotypic, clinical, lifestyle, and genomics data, enhanced the robustness of our analysis. This extensive dataset facilitated the assessment of an extensive range of more than 700 clinical and lifestyle factors, combined with a polygenic risk score (PRS) derived from over 560,000 genetic variants. The integration of these features provided a comprehensive approach to risk prediction, reflecting the multifactorial nature of CAD. Furthermore, the implementation of machine

learning techniques, specifically the stacking ensemble model, enabled us to identify non-linear relationships between the predictive features, further improving the overall performance of the model.

Beyond discrimination, model calibration was assessed using both graphical and statistical measures, including slope, intercept, Brier score and R^2. The model demonstrated strong calibration performance, supporting its suitability for clinical deployment in line with accepted risk communication standards. Additionally, PRC analyses showed that our model outperformed the ACC/AHA score across a range of clinically relevant precision thresholds. This advantage persisted even when stratifying by age deciles, highlighting the robustness of our model across early middle-aged subgroups, a critical consideration given the typically lower event rates in this population. Moreover, DCA confirmed the clinical utility of our model across relevant risk thresholds, while SHAP analysis highlighted the predominant contribution of sex, age, and polygenic risk, supporting both practical applicability and interpretability.

Recent advances in deep learning, particularly the application of temporal models such as Transformer-based architectures, have shown promising potential in capturing dynamic risk trajectories over time for cardiovascular diseases [47, 48] Integrating such temporal models into CAD risk prediction frameworks for early middle-aged adults could offer additional prognostic value by modeling the evolving nature of risk factors [49, 50]. Furthermore, while recent frameworks like QRISK3-Young [27] have tailored traditional risk calculators to younger populations, our approach differs by incorporating both polygenic risk and lifestyle-environmental interactions, offering a more comprehensive assessment of early-onset CAD risk.

While our study demonstrates significant strengths in predicting the 10-year CAD risk among early middle-aged adults, it is essential to address certain limitations. The extensive biobank dataset provided invaluable insights, yet variations in data collection methodologies may have introduced inconsistencies. Additionally, some clinical variables included missing values, requiring imputation strategies that could influence the results. Finally, reliance on self-reported questionnaires for several factors introduces the potential for reporting bias and reduced accuracy.

Conclusions

In conclusion, this study demonstrates the effectiveness of aggregated machine learning approaches in improving CAD risk prediction in early middle-aged adults. By incorporating a broad spectrum of clinical, lifestyle, and genetic factors, our model offers improved discriminative ability and calibration compared to widely used clinical tools. These findings support the utility of integrative predictive frameworks for early detection and prevention. Further research is warranted to validate model performance across diverse cohorts and assess real-world clinical implementation.

Abbreviations

ASCVD	Atherosclerotic cardiovascular disease
CAD	Coronary Artery Disease
PRS	Polygenic risk score
GWAS	Genome-wide association studies
ML	Machine Learning
LD	Linkage disequilibrium
UKB	UK Biobank
GP	General practitioners
SBP	Systolic blood pressure
DBP	Diastolic blood pressure
LDL	Low-density lipoprotein
HDL	High-density lipoprotein
TC	Total cholesterol
PCA	Principal Component Analysis
SNP	Single nucleotide polymorphism
TP	True positives
TN	True negatives
FP	False positives
FN	False negatives
PPV	Positive predictive value
NN	Neuron network, deep learning model
RF	Random Forest, machine learning model
NRI	Net Reclassification Index
PRC	Precision Recall Curve
DCA	Decision Curve Analysis

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Authors' contributions

GN was responsible for manuscript writing and study design. EO was responsible for data analysis, visualization and study design. EF contributed to the methodology, study design and manuscript proofreading. GC contributed to study design and manuscript proofreading. EE contributed to the methodology, study design and manuscript proofreading. All authors have thoroughly examined the manuscript and agree to the publication of this research.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

Erez Ornan and Eran Feldhay are employees of OpenDNA and Georgios Ntritsos has a contract with OpenDNA.

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